

DONGSHENG (DAVID) HAN, Ph.D.

Postdoctoral Scholar, Stanford Bioengineering • NIH/NIA F99/K00 Fellow

Stanford, California • dongshan@stanford.edu • [Google Scholar](#)

RESEARCH PROFILE

Biologist-engineer developing experimental and computational platforms to understand and reprogram cell state. Expertise spans chromatin regulation, nanopore single-molecule genomics, CRISPR perturbation, single-cell analysis, and epigenetic and metabolic strategies for cellular rejuvenation.

RESEARCH EXPERIENCE

NIH/NIA F99/K00 Postdoctoral Scholar | Stanford University **2025–Present**

- Design epigenetic and metabolic interventions to improve T-cell function and persistence in cancer and aging.
- Integrate functional perturbation with multi-omic readouts to identify regulators of T-cell rejuvenation and antitumor activity.

Visiting Ph.D. Researcher | University of California, San Francisco **2024–2025**

- Established direct induced-neuron models that preserve aging signatures and optimized CRISPRa/i delivery to mature neurons.
- Applied single-cell transcriptomic and functional profiling to discover candidate neuronal rejuvenation factors.

Ph.D. Researcher | Vanderbilt University **2020–2025**

- Discovered how the chromatin reader BRWD3 promotes proteasomal degradation of KDM5 to maintain H3K4 methylation homeostasis.
- Developed nanopore-based experimental and analytical methods to map active DNA replication at single-molecule resolution.
- Used ChIP-seq, CUT&RUN, ATAC-seq, biochemistry, and cell biology to define BRWD3 functions in chromatin and replication initiation.

Undergraduate & M.S. Researcher | Sun Yat-sen University **2015–2019**

- Defined developmental functions of the RNA m6A methyltransferase METTL16 using CRISPR-engineered Drosophila models.

EDUCATION

Ph.D., Molecular Biology — Vanderbilt University, 2025

M.S., Biotechnology Engineering — Sun Yat-sen University, 2019

B.S. with Honors, Biology — Sun Yat-sen University, 2017

SELECTED PUBLICATIONS

1. Han D, Schaffner SH, Davies JP, Benton ML, Plate L, Nordman JT. BRWD3 promotes KDM5 degradation to maintain H3K4 methylation levels. Proceedings of the National Academy of Sciences. 2023;120(39):e2305092120. *First author*

2. Han D, Shepherd C, Benton ML, Nordman JT. Nanopore-based sequencing of active DNA replication reveals key principles of metazoan replication fork progression, origin and termination sites. bioRxiv. 2025. *First author; preprint*

3. Han D, Churcher S, Nordman JT. PCR cloning Intermediated Gibson assembly (PIG) for constructing DNA repair templates in CRISPR-Cas9 based gene editing. microPublication Biology. 2023. *First author*

4. Munden A, Wright MT, Han D, Tirgar R, Plate L, Nordman JT. Identification of replication fork-associated proteins in Drosophila embryos and cultured cells using iPOND coupled to quantitative mass spectrometry. Scientific Reports. 2022;12:1–11.

5. Li KK, Han D, Chen F, et al. Compensatory replacement of the BigH1 variant histone by canonical H1 supports normal embryonic development in Drosophila. bioRxiv. 2019.

SELECTED HONORS & FUNDING

- **2024–2029** NIH/NIA F99/K00 Fellowship
- **2024** Graduate Research Excellence Award in Biological Sciences, Vanderbilt University
- **2023** Biomedical Research Education & Training Traveling Grant Award, Vanderbilt University
- **2017** Outstanding Poster Award, National Drosophila Research Conference, China
- **2017** Excellent Undergraduate Thesis and B.S. with Honors, Sun Yat-sen University

TECHNICAL EXPERTISE

Functional genomics	CRISPR-Cas9, CRISPRa/i, perturbation screens, assay development
Single-cell & genomics	10x scRNA-seq, ChIP-seq, CUT&RUN, ATAC-seq, nanopore sequencing
Computation	Python, R, Linux, single-cell analysis, machine learning for genomics
Cell & molecular biology	Primary cell culture, flow cytometry, confocal microscopy, cloning, qPCR, immunoprecipitation, western blot
Research domains	Epigenetics, DNA replication, aging biology, cellular reprogramming, T-cell engineering

LEADERSHIP & COMMUNICATION

Mentored undergraduate, graduate, and research staff trainees; delivered selected departmental and conference talks; served as a Biological Sciences graduate-student DEI representative and international-researcher volunteer.